

AYUSH DAS

Machine Learning Engineer | LLMs & RAG | NLP & Deep Learning
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SUMMARY

Computer Science undergraduate specializing in applied machine learning, with hands-on depth in LLMs, Retrieval-Augmented Generation (RAG), multi-agent systems, and NLP. Build end-to-end ML systems—feature engineering, training, evaluation, deployment—on a foundation of supervised, unsupervised, and deep learning plus probability, statistics, and linear algebra.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, JavaScript

ML & Deep Learning: Supervised & Unsupervised Learning, Neural Networks, PyTorch, TensorFlow, Scikit-Learn, XGBoost, Random Forest, Clustering, Feature Engineering, Model Evaluation

LLMs, NLP & Agents: Natural Language Processing (NLP), Transformers, Hugging Face, DeBERTa-v3, NLI, RAG, Multi-Agent Systems, LangChain, LangGraph, Prompt Engineering, Embeddings, Semantic & Vector Search

Vector & Data: ChromaDB, FAISS, sentence-transformers, Pandas, NumPy

Deployment & MLOps: FastAPI, Streamlit, Server-Sent Events, LangSmith, Git, GitHub, Docker (basics)

Foundations: Probability, Statistics, Linear Algebra, Data Structures & Algorithms

PROJECTS

AI Research Assistant — Autonomous Multi-Agent RAG Pipeline | [GitHub](#)

Jun 2026

- Architected a 5-agent autonomous research system on a **LangGraph StateGraph** (Planner, Searcher, Reader, Critic, Writer) with conditional routing and a self-critique retry loop that re-searches detected coverage gaps before generating a sourced report.
- Engineered **dual-layer memory**—typed short-term graph state plus a long-term **ChromaDB** vector store with sentence-transformer embeddings—so each run retrieves semantically similar prior research and improves across sessions.
- Wired a tool-calling layer (Tavily search + BeautifulSoup extraction) and streamed live agent execution to the browser over **FastAPI + Server-Sent Events**; instrumented every step with LangSmith tracing, and hardened the run with graceful degradation (JSON retries, failed-URL skips, memory-outage fallbacks) so no single failure crashes it.

Legal Clause Conflict Resolver — Explainable Legal NLP | [GitHub](#)

May 2026

- Fine-tuned **DeBERTa-v3-large** for Natural Language Inference to detect contradictions between clause pairs in SEC EDGAR filings, reaching **>85% validation accuracy** and cutting false positives ~15% over the baseline through threshold tuning; aligned related clauses with sentence-transformers and FAISS to reduce redundant inference.
- Delivered **explainable AI** via cross-attention heatmaps that surface the exact tokens driving each contradiction, plus a severity scorer that ranks the highest-risk conflicts first; served through a FastAPI backend and analytics dashboard.

YT Genius — RAG Chatbot over YouTube Transcripts | [GitHub](#)

Dec 2025

- Developed an end-to-end **Retrieval-Augmented Generation** assistant that ingests YouTube transcripts and answers grounded questions, retrieving relevant context before generation to curb hallucination.
- Implemented retrieval with **FAISS** vector search and LangChain over an OpenAI LLM, served via a FastAPI backend and a React/Vite frontend.

Library DNA Discovery — Unsupervised Clustering at Scale | [GitHub](#)

May 2026

- Engineered a two-stage clustering pipeline over **487K titles and 2.3M circulation records**, achieving a **0.78 silhouette score** and 0.48 Davies-Bouldin index—strong separation and cluster stability.

EXPERIENCE

VanillaCart — Web Development Intern

Aug 2025 – Oct 2025

- Rebuilt a client website into an improved, fully responsive version with HTML, CSS, and JavaScript, modernizing layout and structure for a cleaner user experience.
- Applied on-page **SEO best practices**—semantic markup, metadata, and optimized assets—improving search visibility and organic traffic; managed content migration and collaborated with developers to deploy and maintain production-ready sites.

CERTIFICATIONS

[IBM – Python for Data Science & AI, Completed June 2025](#) • [CipherSchools – Machine Learning with Data Science, Completed July 2025](#)
• [IBM – ChatGPT-4 Prompt Engineering, Completed August 2025](#) • [Google – Prompt Engineering Essentials, Completed June 2025](#) • [Udemy – Generative AI & Tools, Completed August 2025](#)

EDUCATION

B.Tech, Computer Science & Engineering

Aug 2023 – Present

Lovely Professional University, Punjab, India — CGPA: 8.08 / 10